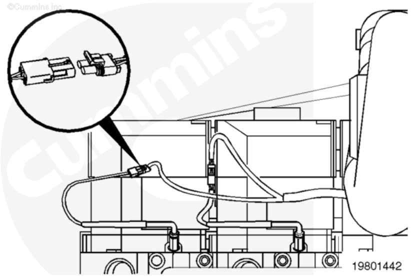
Trainings

Remove

Note : In order to gain access to the exhaust temperature sensor wires for their removal, it may be necessary to remove other bracketry and/or other support or anchoring hardware for engine components such as: the low temperature aftercooling (LTA) coolant tube (remove coolant first), coolant hose, and turbocharger oil supply line.

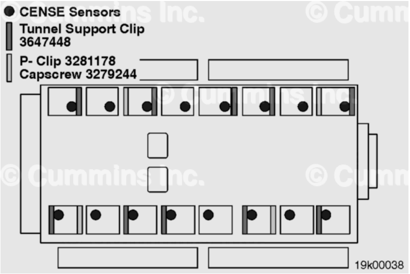


Lift up on the locking tab and pull the electrical connectors apart.

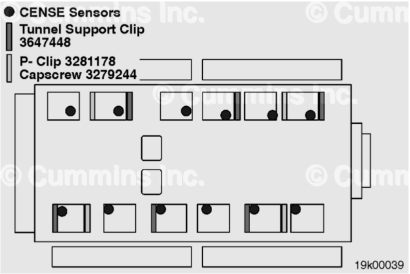
Remove the sensor from the cylinder head.

The sensor wires are secured to the engine by wiring harness tunnel support clips and P-clips.

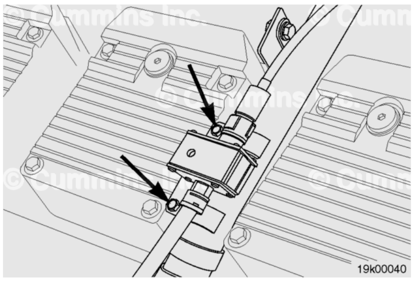
QSK60 engines have five tunnel support clips and two P-clips per bank.



QSK45 engines have three tunnel support clips and two P-clips per bank.

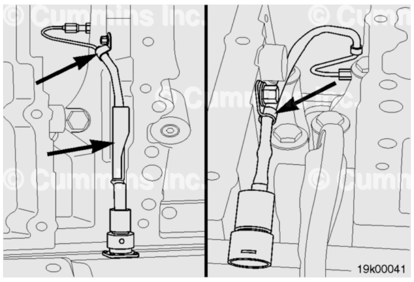


Loosen the capscrews that clamp down the sensor to the CENSE™ exhaust gas temperature sensor harness rail.



Note : Certain tunnel support clips and/or P-clips might need to be removed during this step.

Remove the sensor wires from the engine.



Install

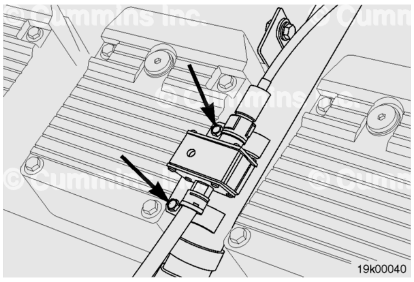
Apply anti-seize compound to the sensor threads.

Install the sensor onto the engine.

Note : The connector area of the sensor is susceptible to heat damage. Make sure that the sensor connector assembly does not contact the exhaust manifolds or turbochargers.

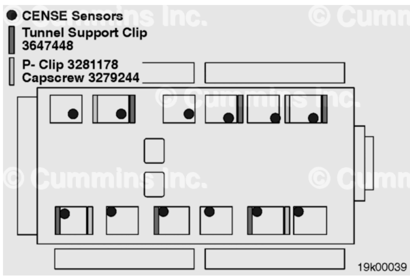
Tighten the capscrew that secures the sensor to the exhaust gas temperature sensor harness rails.

Torque Value: 24 n•m [212 in-lb]



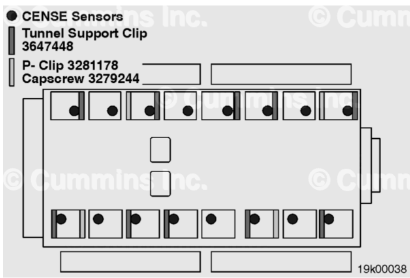
Note : Do not stretch the sensor wire as you route it through the brackets, clamps, and P-clips. Do not crush the sensor wire by over tightening the capscrew nuts.

Route the QSK45 exhaust gas temperature sensor wiring per the hardware schematic shown in this step. Note that each sensor will be routed by each individual cylinder head and then plugged into the harness in between the valve covers and the aftercooler housing



Note : Do not stretch the sensor wire as you route it through the brackets, clamps, and P-clips. Do not crush the sensor wire by over tightening the capscrew nuts.

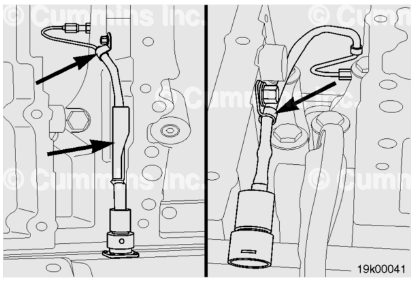
Route the QSK60 exhaust gas temperature sensor wiring per the hardware schematic shown in this step. Note that each sensor will be routed by each individual cylinder head and then plugged into the harness in between the valve covers and the aftercooler housing.



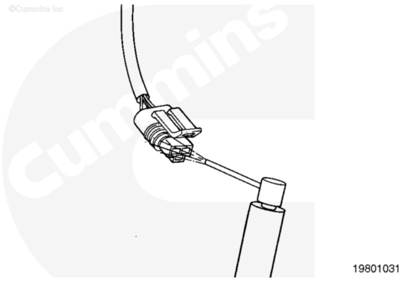
Install the sensor wire through the P-clips and the tunnel support clips as shown in the QSK60 or QSK45 schematics shown in the previous steps. Tighten the capscrews to the specified torque value.

Note : Retrace the route of the sensor wire to verify that it is not in close proximity to a heat source or touching any other engine component that can damage or wear the sensor wire, causing premature sensor malfunction.

Torque Value: 15 n•m [133 in-lb]

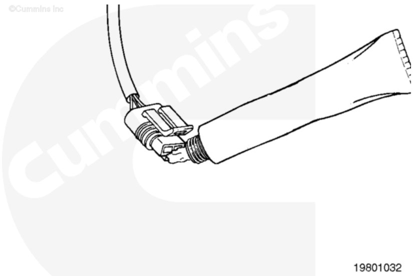


Use electrical contact cleaner, Part Number 3824510, to remove all dirt and moisture from the sensor connector port and the harness connector.



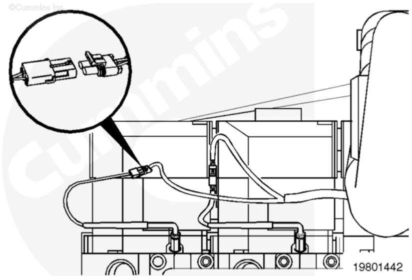
⚠ CAUTION ⚠

Use only Cummins-recommended lubricant DS-ES, Part Nunmber 3822934. Other lubricants, such as lubricating oil or grease, in the connectors can cause engine control module (ECM) damage, poor engine performance, or premature connector wear.



Apply a small amount of lubricant to the connector terminals. Before installing, fill the entire connector cavity with lubricant.

Push the sensor and sensor harness connectors together, until they lock.



Last Modified: 31-Oct-2013